Abstract linear Algebra homwork4

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Problem1

(a) True

(b) Not true

Because if we consider a complex coefficient polynomial$(1+i)x$ $(1-i)x$, we can check the linearity of the linear mapping. So $T(x+y)$ is actually $(x+y)^2$. For the polynomial we have give , we have $((1+i)x+(1-i)x)^2 = 4x^2$. But for $T(1+i)(x)+T(1-i)(x) = 2ix^2-2ix^2 = 0$ which means the map do not satisfy the linearity.

(c) False

Because V is actually infinite-dimensional and the basis should be infinite either

Problem2

Let $V = \mathbb{R}[x] \leq n$ with ordered basis $\mathcal{B}(1, x^1, x^2, ..., x^n)$

(a)

Proof. For any $g, f \in V$ and $c, d \in \mathbb{R}$

$$\varphi(cf + dg) = \int_0^1 (cf(t) + dg(t))dt = c\int_0^1 f(t)dt + d\int_0^1 g(t)dt = c\varphi(f) + d\varphi(g).$$

□

Hence φ is $\mathbb{R}$-linear, so $\varphi \in V^*$ Let $\mathcal{B}^* = (e_0^*, e_1^*, ..., e_n^*)$ be the dual basis, where $e_k^*(x^j) = \delta_{kj}$. Evaluating φ on each basis vector gives

$$\varphi(x^k) = \int_0^1 t^k dt = \frac{1}{k+1}, 0 \leq k \leq n$$

Therefore

$$\varphi = \sum_{k=0}^{n} \frac{1}{k+1} e_k^* = e_0^* + \frac{1}{2}e_1^* + ... + \frac{1}{n+1}e_n^*.$$

(b)(n=4)

We compute the basis such that the result in $\mathcal{B}$ is

$$T(1) = 0, T(x) = 1, T(x^2) = 2x, T(x^3) = 3x^2, T(x^4) = 4x^3$$

Writing these as column vectors in $\mathcal{B}$, the matrix of T is

$$[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

For the dual map $T^* : V^* \to V^*$, recall $T^*(\psi) = \psi \circ T$. Let $B^* = (f_0, f_1, f_2, f_3, f_4)$ with $f_i(x^j) = \delta_{ij}$. We evaluate:

$$\big(T^*(f_i)\big)(x^j) = f_i\big(T(x^j)\big) = f_i(j\, x^{j-1}) = j\, \delta_{i,\, j-1}.$$

This is nonzero only when $j = i + 1$, where it equals $i + 1$. Hence:

$$T^*(f_0) = f_1, \quad T^*(f_1) = 2f_2, \quad T^*(f_2) = 3f_3, \quad T^*(f_3) = 4f_4, \quad T^*(f_4) = 0.$$

The matrix of T^* with respect to B^* is therefore

$$[T^*]_{B^*} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \end{pmatrix}.$$

Indeed, $[T^*]_{B^*} = \big([T]_B\big)^\top$, as required.

Problem3

(a)

So we can easily obsevre that $\mathbb{R}$ is a subfield of $\mathbb{C}$. Since all the complex number has the form $a + bi$, then we can easily find the basis is actuallly $\{1, i\}$, so the dimension is 2.

(b)

Let V be a vector space over $\mathbb{C}$. We explain why V is also a vector space over $\mathbb{R}$.

V already carries an abelian group structure $(V, +)$ and a $\mathbb{C}$-scalar multiplication satisfying all vector-space axioms. Since $\mathbb{R}$ is a subfield of $\mathbb{C}$, every real number $r \in \mathbb{R}$ can be viewed as an element of $\mathbb{C}$. Define $\mathbb{R}$-scalar multiplication by restriction: for $r \in \mathbb{R}$ and $v \in V$, set $r \cdot v$ to be the existing $\mathbb{C}$-scalar product of r (regarded as a complex number) with v. All eight vector-space axioms are inherited from the $\mathbb{C}$-scalar multiplication. Hence V is an $\mathbb{R}$-vector space.

(c)

Claim. *If V is a finite-dimensional $\mathbb{C}$-vector space, then* $\dim_{\mathbb{R}}(V) = 2 \cdot \dim_{\mathbb{C}}(V)$.

Proof. Let $n = \dim_{\mathbb{C}}(V)$ and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V over $\mathbb{C}$. We claim that

$$\mathcal{B}_{\mathbb{R}} = (v_1, \ldots, v_n,\ iv_1, \ldots, iv_n)$$

is a basis for V over $\mathbb{R}$.

Spanning. Let $v \in V$. Since $\mathcal{B}$ is a $\mathbb{C}$-basis, write $v = \sum_{j=1}^{n} z_j v_j$ with $z_j = a_j + b_j i$, $a_j, b_j \in \mathbb{R}$. Then

$$v = \sum_{j=1}^{n} (a_j + ib_j) v_j = \sum_{j=1}^{n} a_j\, v_j + \sum_{j=1}^{n} b_j\, (iv_j),$$

which is an $\mathbb{R}$-linear combination of elements of $\mathcal{B}_{\mathbb{R}}$.

Linear independence. Suppose

$$\sum_{j=1}^{n} a_j\, v_j + \sum_{j=1}^{n} b_j\, (iv_j) = 0, \quad a_j, b_j \in \mathbb{R}.$$

Rewriting, $\sum_{j=1}^{n} (a_j + ib_j)\, v_j = 0$. Since $(v_1, \ldots, v_n)$ is a $\mathbb{C}$-basis and each coefficient $a_j + ib_j \in \mathbb{C}$, we conclude $a_j + ib_j = 0$ for all j, i.e., $a_j = b_j = 0$.

Therefore $\mathcal{B}_{\mathbb{R}}$ is a basis for V over $\mathbb{R}$, and $\dim_{\mathbb{R}}(V) = 2n = 2 \cdot \dim_{\mathbb{C}}(V)$. □

Problem5

Claim. *If $T\colon V \to V$ has the same matrix with respect to every ordered basis of V, then $T = \lambda id_V$ for some $\lambda \in \mathbb{F}$.*

Proof. Let $n = \dim V$. Fix an ordered basis $\mathcal{B} = (e_1, \ldots, e_n)$ and let $A = [T]_{\mathcal{B}}$. For any other ordered basis $\mathcal{B}'$, the change-of-basis formula gives $[T]_{\mathcal{B}'} = P^{-1}AP$, where P is the invertible change-of-basis matrix. By hypothesis $[T]_{\mathcal{B}} = [T]_{\mathcal{B}'}$, so

$$P^{-1}AP = A \quad \Longleftrightarrow \quad AP = PA,$$

for every invertible $n \times n$ matrix P over F (every such P is a change-of-basis matrix for some basis $\mathcal{B}'$).

We now show $A = \lambda I_n$. For $i \neq j$, let E_{ij} be the matrix with 1 in position (i, j) and 0 elsewhere, and set $P_{ij} = I + E_{ij}$. This is invertible (it is an elementary matrix with $\det(P_{ij}) = 1$), so $AP_{ij} = P_{ij}A$, giving

$$AE_{ij} = E_{ij}A. \tag{1}$$

We compute both sides entry-wise. For any indices k, ℓ:

$$(AE_{ij})_{k\ell} = \sum_{m} A_{km}(E_{ij})_{m\ell} = A_{ki}\, \delta_{j\ell},$$

$$(E_{ij}A)_{k\ell} = \sum_{m} (E_{ij})_{km} A_{m\ell} = \delta_{ki}\, A_{j\ell}.$$

Equation (1) therefore gives $A_{ki}\, \delta_{j\ell} = \delta_{ki}\, A_{j\ell}$ for all k, ℓ.

- *Setting $\ell = j$:* $A_{ki} = \delta_{ki}\, A_{jj}$. For $k \neq i$ this yields $A_{ki} = 0$; for $k = i$ this yields $A_{ii} = A_{jj}$.

Since this holds for every pair $i \neq j$, every off-diagonal entry of A is zero and all diagonal entries are equal to some $\lambda \in F$. Hence $A = \lambda I_n$, i.e., $T = \lambda\, \mathrm{id}_V$. □

Problem6

(a)

First we claim that, once we choose a basis for a linear map, the linear map is actually isomorhpic to the matrix. So for instance, the invertible linear maps refer to the invertible matrix. Since for each linear map, the matrix is unique. Thus the number of invertible linear map is the number of invertible matrix.

Now we consider for $V \to V$ with the dimension n, the matrix should be $n \times n$ matrix. And the element should be in the field $\mathbb{F}_p$. If a matirx is an invertible matrix, consider it's column vectors should all be linearly independent, that is to say, no such two column vectors in this matrix, one is the multiplier of another.

- **Column 1** (call it c_1): must be nonzero. There are $p^n - 1$ choices.
- **Column 2** (c_2): must not lie in span$\{c_1\}$. The subspace span$\{c_1\}$ has $p^1 = p$ elements, so there are $p^n - p$ choices.
- **Column** k (c_k): must not lie in span$\{c_1, \ldots, c_{k-1}\}$. Since $c_1, \ldots, c_{k-1}$ are linearly independent, their span has p^{k-1} elements, leaving $p^n - p^{k-1}$ choices.

By the multiplication principle, the number of invertible linear maps $V \to V$ equals the number of invertible $n \times n$ matrices over $\mathbb{F}_p$, which is

$$\prod_{k=0}^{n-1}(p^n - p^k) = (p^n - 1)(p^n - p)(p^n - p^2)\cdots(p^n - p^{n-1}).$$

(b)

We have already proof the statement in the subsection(a)

Problem12

Recall the Rank-Nullity Theorem

$$null(T) + rank(T) = dim(v)$$

From the condition, we can change the basis of null space into$(5x_2, x_2, 7x_4, x_4)$, such that the basis for null space:

$$\{(5, 1, 0, 0)(0, 0, 7, 1)\}$$

which means $null(T) = 2$, since $dim(F^4) = 4$ such that according to rank-nullity theorem

$$rank(T) = 4 - 2 = 2 = dim(F^2)$$

and then T is surjective.

Check-In

I want to have a big meal after midterm!